

0G Validator Layer Scope 2 Emissions Methodology – Mainnet (MiCA-Aligned)



1) Scope & Purpose

This document defines a MiCA-aligned methodology to estimate Scope 2 emissions (electricity) for the validator layer of 0G on mainnet, with a testnet case study based on real validator locations and power profiles captured earlier in this thread. It mirrors the format of the storage methodology and is designed for repeatable reporting and regulator review.

Boundary: All validator nodes running the software, including those in the active set (participating in consensus) and those not in the active set (e.g., staked candidates or inactive validators maintaining sync). Storage, DA, compute, and serving layers are out of scope here.

2) Validator Hardware

Component	Mainnet	Testnet
Memory	64 GB	64 GB
CPU	8 cores	8 cores
Disk	1 TB NVMe SSD	4 TB NVMe SSD
Bandwidth	100 Mbps up/down	100 Mbps up/down

Power model:

- Testnet: CPU+RAM+NIC+misc $\approx$ 188 W IT; NVMe (4 TB) $\approx$ 11 W IT $\rightarrow$ 199 W IT. PUE $\approx$ 1.4 $\rightarrow$ $\sim$ 280 W wall.
- Mainnet: 1 TB NVMe $\approx$ 3 W IT vs 11 W for 4 TB $\rightarrow$ -8 W IT. With PUE $\approx$ 1.4 $\rightarrow$ -11.2 W wall. Result: $\sim$ 269 W wall.

Note: Empirical wall power measurement will be conducted using our own validator hardware post-mainnet launch to validate these modeled values.

3) Calculation Method

Let $N(d)$ = online validators (active + powered standby + powered inactive/candidate) on day d .

1. Total network power (W):

$$P_{\text{net}}(d) = N(d) * P_{\text{node}}$$

2. Daily energy (kWh):

$$E_{\text{day}}(d) = (P_{\text{net}}(d) / 1000) * 24$$

3. Annual energy (kWh/year):

$$E_{\text{yr}} = \text{sum_over_days}(E_{\text{day}}(d))$$

4. Location-based emissions (kg CO₂/year):

$$\text{CO2}_{\text{yr}} = \text{sum_over_validators}((P_{\text{node}}/1000) * 8760 * EF_{\text{c}}(i))$$

Where unknown locations use IEA world average $EF = 0.459$ kg CO₂/kWh.

5. Convert to metric tons:

$$\text{tCO2}_{\text{yr}} = \text{CO2}_{\text{yr}} / 1000$$

Legend:

- P_{node} : Avg wall power per validator (W), incl. PUE
- EF_{c} : Grid emission factor (kg CO₂/kWh) for country c
- PUE: Power Usage Effectiveness

4) Testnet Case Study (63 Nodes, Location-Weighted)

Using the 37-node location dataset's average per-node energy and weighted EF, scaled to 63 nodes:

- Per-node annual energy: ~2,306.27 kWh/yr
- Network annual energy: ~145,295 kWh/yr
- Weighted EF: ~0.375 kg CO₂/kWh
- Annual emissions: 54.5 tCO₂/yr

Reconciliation with earlier ~47.5 tCO₂/yr estimate

The earlier ~47 tCO₂/yr figure used a more EU-leaning emission factor (~0.32 kg/kWh) and a slightly different per-node energy assumption (~2,356 kWh/yr). The current case study applies the location-weighted EF (~0.375 kg/kWh) derived from the 37-node mix and a 2,306 kWh/yr per-node average, then scales to 63 nodes. The methodology now documents these inputs explicitly so changes in EF or per-node energy transparently explain differences in totals.

5) Mainnet Projection (200 Validators)

A) Global-average EF (0.459):

Per-node energy = $(269/1000) \times 8760 = 2,356.44$ kWh/yr
Network energy = $2,356.44 \times 200 = 471,288$ kWh/yr
CO₂ (t/yr) = $(471,288 \times 0.459) / 1000 = 216.2$ tCO₂/yr

B) Location-weighted EF (~0.375):

CO₂ (t/yr) = $(471,288 \times 0.375) / 1000 = 176.7$ tCO₂/yr

6) CCRI/Solana Methodology Alignment

- **Node Power Measurement:** Commitment to empirical wall power measurement for at least one active and one standby validator, similar to CCRI's multi-configuration

approach.

- **Inactive Validators:** Includes powered inactive/candidate validators in N(d) to meet MiCA's comprehensive accounting.
- **Geographic EF:** Matches CCRI in applying location-weighted EF and uses IEA's 0.459 kg CO₂/kWh as fallback.
- **TPS Adjustment:** Added to the mainnet roadmap post-launch, measuring marginal energy per transaction.
- **Hardware Transparency:** Will publish hardware measurement table, replacing modeled values with empirical ones over time.

7) Improving Accuracy

- Daily IP geolocation for EF weighting
- Empirical wall power for representative nodes
- Track standby/inactive node utilization
- Update P_node when hardware specs change
- Integrate TPS vs. power regression post-mainnet

Summary:

- 63-node testnet (location-weighted): ~54.5 tCO₂/yr
- Projected 200-validator mainnet: ~176–216 tCO₂/yr
- MiCA-aligned, CCRI-informed, and ready for live monitoring integration

Year one estimates:

0G Validator Layer — Year-1 Scope 2 (MiCA-aligned)

Inputs

- Validators: 121 (all running)
- Per-node wall power: 269 W $\rightarrow$ 2,356.44 kWh/year per node
- Method: location-based EF where known; unknowns set to non-EU EF = 0.55 kg CO₂/kWh (conservative)

Location mix used (64 known from your list)

Europe (incl. UK/UA) 40 • North America 9 • Japan 4 • Korea 4 • China 2 • Singapore 2 • Turkey 2 • Australia 1

(57 validators unknown $\rightarrow$ treated as non-EU at EF 0.55)

Emission factors (kg CO₂/kWh)

EU 0.30 • NA 0.40 • JP 0.46 • KR 0.50 • CN 0.65 • SG 0.52 • TR 0.47 • AU 0.70 • Unknown (non-EU) 0.55

Results

- Total energy (121 nodes): $121 \times 2,356.44 = 285,129.24$ kWh/year
- Known 64 validators: 55.2 tCO₂/year
- Unknown 57 validators (EF 0.55): 73.9 tCO₂/year
- Total (conservative): 129.1 tCO₂/year

Sense-check / fallback

Using global-average EF 0.459 for all 121 nodes would yield 130.9 tCO₂/year. This approach is similarly conservative and avoids assigning unknowns to low-EF EU grids.